

Single-frequency model

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$$u = \frac{t_1 + t_2}{2}, \quad v = t_2 - t_1$$
$$C(t_1, t_2) = C_0 + A \cos(\omega v) + m \cos(\omega u + \phi) \cos\left(\frac{\omega}{2} v\right)$$

Overview

A clean way to think about this model physically is to separate **what fluctuates**, where it **fluctuates**, and **how those fluctuations are coupled in time**.

1. The $A \cos(\omega v)$ term: stationary oscillatory dynamics

This term depends **only on the lag time v** .

Physically, this corresponds to:

- A *stationary*, time-translation-invariant process
- A well-defined oscillatory mode with period $T = 2\pi/\omega$
- Something like a coherent relaxation, hopping, or vibrational process that is always present

In XPCS language, this is what you would expect if:

- A subset of scatterers undergoes periodic or quasi-periodic dynamics
- The dynamics do not age or drift during the measurement

This is the “safe” term, it behaves like what g_2 -based intuition expects.

2. The $m \cos(\omega u + \phi)$ factor: non-stationarity or slow modulation

This part depends on **the mean time u** .

That means:

- The dynamics are not the same at early and late times
- The system is being slowly modulated, intermittently activated, or drifting

Physically, this can represent:

- Slow environmental or structural evolution
- Periodic activation of dynamics (e.g. stress build-up and release)
- Switching between dynamical states

In a Bragg XPCS context, this is exactly what you would expect from:

- Grain-boundary rearrangements
- Intermittent defect motion
- Phase-front breathing or periodic strain accumulation

This term alone would already violate stationarity, hence the TTC structure.

3. The $\cos(\frac{\omega}{2} v)$ factor: subharmonic lag-time structure

This is the most interesting part.

Compared to the main A-term:

- The lag-time oscillation is **slower by a factor of 2**
- The system “remembers” correlations over longer lag times

Physically, this suggests:

- A **two-step or paired process**
- Correlations that require *two events* to decorrelate fully
- Alternating forward-backward motion, stick-slip, or reversible rearrangements

This is very natural for:

- Periodic grain-boundary diffusion
- Back-and-forth defect migration
- Elastic loading/unloading cycles

Why the half-frequency term?

Yes – it’s important, and it’s not an arbitrary choice. A **half-frequency in the v direction is a very strong physical signature**. Here’s why it shows up and what it usually means.

1. What “half frequency in v ” really means

Your term is $\cos(\frac{\omega}{2} v)$, $v = t_2 - t_1$

That means:

- The **correlation repeats only after twice the period** of the underlying oscillation
- The system needs **two cycles** before it “looks the same again” in lag time

This is classic **period doubling in correlations**, not necessarily in the motion itself.

2. Why correlations can have half the frequency of the motion

Suppose a microscopic displacement is:
 $x(t) = x_0 \cos(\omega t)$

The *scattering phase* depends on displacement squared or on phase differences:
 $\Delta \phi \sim x(t_2) - x(t_1)$

Now consider what happens over **one oscillation**:

- After half a cycle, the displacement reverses sign
- The structure may be in a *different configuration*, but not fully decorrelated
- Only after a **full forward-backward cycle** does the phase difference repeat

So the correlation function naturally oscillates at:
 $\omega_{\text{corr}} = \frac{1}{2} \omega_{\text{motion}}$

This is extremely common in:

- Reversible motion
- Elastic deformation
- Back-and-forth defect hopping

3. Physical picture: two-state or back-and-forth dynamics

A half-frequency almost always means **the system has two equivalent configurations**.

Examples:

- Grain boundary moves left $\rightarrow$ right $\rightarrow$ left
- Defect hops A $\rightarrow$ B $\rightarrow$ A
- Strain loads $\rightarrow$ unloads $\rightarrow$ loads

The *microscopic motion* has frequency ω , but the *configuration* repeats every **two steps**, so correlations repeat at $\omega/2$.

This is why the TTC shows:

- Diamonds or chequerboards
- Alternating bright/dark lobes
- Strong anti-diagonal structure

4. Why this matters for Bragg XPCS (especially)

At a Bragg peak:

- Intensity is sensitive to **phase**, not just magnitude
- Opposite displacements can produce **similar intensities**

4. Why the product structure matters

The product $\cos(\omega u)\cos(\frac{\omega}{2}v)$ means:

The lag-time correlations only appear strongly during certain global-time windows.

This produces:

- Diamond or chequerboard TTC patterns
- Oscillations that are *localized* in u
- Masks where oscillations appear and disappear

This matches your observation that:

- Only specific masks around the Bragg peak show oscillations
- The diagonal is not always the strongest feature
- The TTC shows repeating but localized structure

5. Physical interpretation in one sentence

A concise physical picture is:

A periodically active nanoscale process (grain boundary, defect cluster, strain front) undergoes coherent oscillatory motion, but only during certain phases of a slower cycle, producing non-stationary, subharmonic correlations in the TTC.

6. Why this is especially compelling for Bragg XPCS

At a Bragg peak:

- You are sensitive to **phase and strain**, not just density
- Small reversible displacements can dominate the signal
- Collective motion of ordered regions naturally produces oscillatory TTC features

Your model is therefore not “toy” in a dismissive sense, it is **the minimal mathematical structure** that captures:

- Stationary dynamics
- Non-stationarity
- Mode coupling
- Subharmonic temporal structure

- Sign changes in displacement don’t immediately decorrelate the speckle

So Bragg XPCS is *especially prone* to half-frequency effects.

In diffuse scattering you might lose this; in Bragg it survives.

5. Why the half-frequency appears only in the modulated term

Your model structure is:

$$A\cos(\omega v) \cos(\omega u)\cos\left(\frac{\omega}{2}v\right)$$

Interpretation:

- The **stationary background** dynamics decorrelate every cycle → full frequency
- The **intermittent / gated dynamics** require two cycles → half frequency

That tells you these are **not the same physical process**.

This is a huge clue experimentally.

6. Experimental implications (very concrete)

If the half-frequency is real (not a fitting artefact), then:

- The dynamics are **reversible or near-reversible**
- You are seeing **collective motion**, not random diffusion
- Grain boundaries or strain fields are far more likely than ionic hopping
- The system is **not ergodic on the measurement timescale**

This also explains:

- Why only some masks show oscillations
- Why TTC diagonals aren’t always maximal
- Why g_z alone hides the physics

7. One-sentence takeaway

A half-frequency in v means the system must complete a full forward-and-back cycle before its microscopic configuration truly repeats, which is the hallmark of reversible, collective dynamics such as elastic or grain-boundary motion.